Lecture 14
Time Series Data and Markov Models

Algorithms, AI & Data Science II

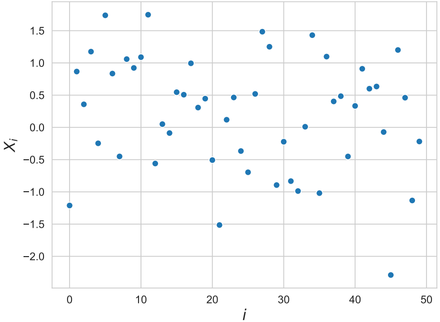
Prof. Dr. Ingo Scholtes
Chair of Machine Learning for Complex Networks
Center for Artificial Intelligence and Data Science (CAIDAS)
Julius-Maximilians-Universität Würzburg
ingo.scholtes@uni-wuerzburg.de

Notes:

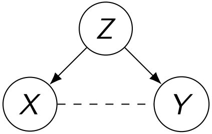
- **Lecture L14:** Time Series Data and Markov Models 25.06.2025
- **Educational Objective:** We discuss statistical models for time series data and introduce Markov chains to model categorical sequences. Exploring random walks in graphs, we clarify under which conditions Markov chains reach a stationary state.
  - Stochastic Processes and Markov chains
  - Random Walks in Graphs
  - Stationary States of Markov Chains
- **Handout version:** *2025/06/23 10:18:06*

Motivation

- we introduced techniques for **statistical modelling and inference** in empirical data
- we considered causal diagrams, where nodes are **random variable** and edges represent **causal relationships or correlations**
- example for **probabilistic graphical model** that can be used to represent **relationships in data**
- so far, we considered sets of **independent realizations** X_i of random variable
- how can we model **time series data**, i.e. values X_t recorded at different times t



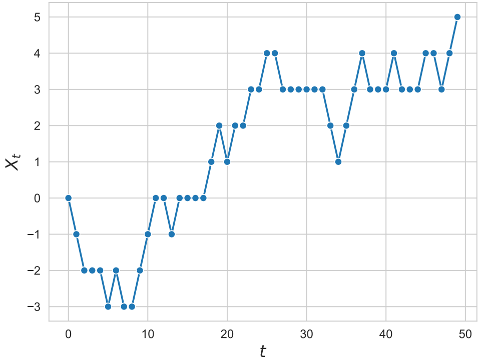
50 independent realization X_i of random variable $X \sim$ Norm



Notes:

Stochastic processes

- consider **time series data** X_t with time index t
- we call collection of random variables $X_t \in E$ a **stochastic process** with state space E
- we can have **discrete or continuous state space**, e.g. $X_t \in \mathbb{R}$ or $X_t \in \{a, b, c\}$
- we can have **discrete or continuous time**, e.g. $t \in [0, \infty)$ or $t \in \{0, 1, 2, \ldots\}$
- how can we **model patterns** in such stochastic processes?
- we can **use probabilities to model uncertainty about future states** X_t of the process

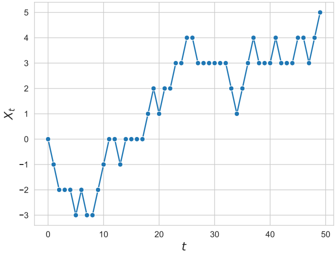


discrete-time stochastic process with **discrete state space**

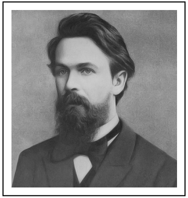
Notes:

Markov models

- we can use **probabilities** to model **stochastic processes** and resulting **time series data**
- assign $P(X_t)$ of **future states**, possibly conditional on current/past states
- Markov models assume that system **memoryless**, i.e. next state only depends on current state and not on past states
  - continuous-time $\Rightarrow$ **Markov process**
  - discrete-time $\Rightarrow$ **Markov chain**
- important foundation for **stochastic modelling of time series data**
- numerous applications in **pattern recognition, time series forecasting, recommender systems**



discrete-time process with **discrete state space**



Andrey Markov
1856 – 1922

image credit: Wikimedia Commons, public domain

Notes:

Markov chains

- consider **discrete-time** stochastic process on **finite, discrete state space** where X_t is state of process at time $t \in \{0, 1, \dots\}$
- $P(X_t = s)$ models uncertainty about state $s \in E$ of process at time t
- at time t **transition probability** from state s_t to state s_{t+1} can be given as

$$P(X_{t+1} = s_{t+1} | X_t = s_t, X_{t-1} = s_{t-1}, \dots X_0 = s_0)$$

- process is **memoryless**, i.e. has the **Markov property**, iff

$$P(X_{t+1} = s_{t+1} | X_t = s_t) = P(X_{t+1} = s_{t+1} | X_t = s_t, X_{t-1} = s_{t-1}, \dots X_0 = s_0)$$

  for all sequences of states $s_0, \dots, s_{t-1} \in E$
- we obtain a **Markov chain model** with discrete and finite state space

observations

- next state X_{t+1} only depends on current state X_t
- transition probabilities between states s_i and s_j can be given in entries T_{ij} of **transition matrix** $\mathbf{T} \in \mathbb{R}^{n \times n}$ where $n = |E|$

Notes:

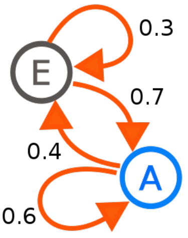
Applications of Markov chains

- Markov chains are important class of **probabilistic graphical models**
- nodes of graph represent **states of a system**, while edges represent **transitions between states**
- assuming model for transitions, Markov chains allow to answer questions about **dynamic behavior of systems**

exemplary questions

- what is the **probability that system is in given state** at given point in time?
- how long will it take on average **to first reach a given state**?
- does system reach a **stationary state** in which the probabilities of states seize to change?

- **important applications across disciplines**



graph representation of Markov chain with two states A and E

image credit: Wikimedia Commons, User Joxemai4, CC-BY-SA

exemplary applications of Markov chains

- packet arrivals in communication networks
- user click streams in information systems
- diffusion processes in networks
- search and rescue operations

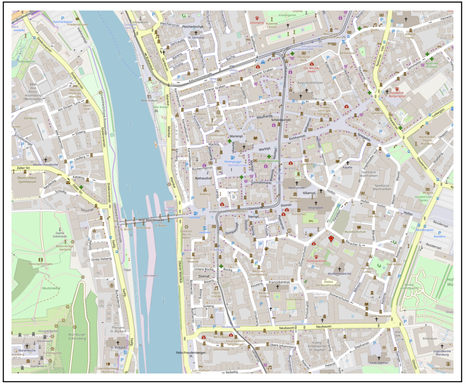
Notes:

Example application

exemplary "search and rescue" application

- you loose your drunk friend in the streets of Würzburg
- you need to search for him/her?

- **model Würzburg as graph** $G = (V, E)$, where nodes V represent "locations" and (undirected) edges $E \subseteq V \times V$ represent connecting streets
- we call node sequence $v_0, v_1, \dots, v_t$ with $(v_i, v_{i+1}) \in E \quad (i = 0, 1, \dots, t-1)$ **walk in graph** G
- you know **starting node** but you do not know which walk your friend has taken in the graph
- idea: use a Markov chain to model **lack of knowledge** about current location (i.e. node) of your friend



Map of Würzburg

image credit: Map created from openstreetmap data

Notes:

Random walks in graphs

- we can use Markov chain to model **random walk in graph** $G = (V, E)$ with nodes V and edges $E \subseteq V \times V$

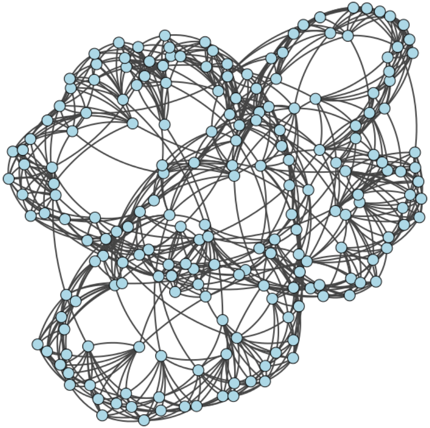
random walk as Markov chain

For a graph $G = (V, E)$, consider Markov chain model for random walks with discrete **state space** $E = V$.

- random variable $X_t \in V$ assumes state of the walker (i.e. visited node) at time $t \in \{0, 1, 2, \ldots\}$
- transitions must (generally) follow edges of graph, i.e. $P(X_t = v_t | X_{t-1} = v_{t-1}) > 0 \implies (v_{t-1}, v_t) \in E$
- any realization of the stochastic process X_t is a walk through graph G

- important class of models for **diffusion processes in networks** $\longrightarrow$ Course: Statistical Network Analysis
- basis for **machine learning and recommender systems**

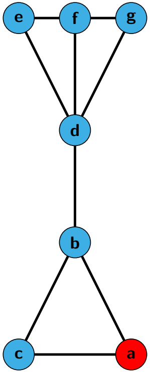
  $\longrightarrow$ Course: Machine Learning for Complex Networks



random walk in an example graph

Notes:

Adjacency matrices of graph



random walker with initial state $X_0 = a$

how can we define **transition matrix T of Markov chain** modelling random walk in G?

we can use entries A_{ij} of **adjacency matrix** $\mathbf{A}$ to represent graph $G = (V, E)$, i.e.

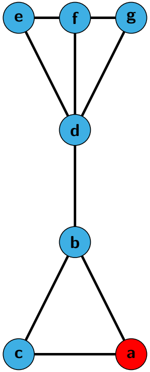
$$A_{ij} = \begin{cases} 1 \text{ iff } (i,j) \in E \\ 0 \text{ else} \end{cases}$$

where i refers to row and j refers to column

$$\mathbf{A} = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} \begin{array}{c} \begin{array}{ccccccc} a & b & c & d & e & f & g \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Notes:

Transition matrix of random walk



random walker with initial state $X_0 = a$

how else could we define **transition probabilities**?

we can **define entries of transition matrix** as

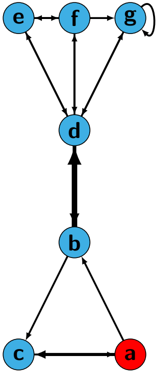
$$T_{ij} := \frac{A_{ij}}{\sum_{k \in V} A_{ik}} = \frac{1}{d_i}$$

where T_{ij} is **probability of walker** in node i to move to node j and d_i is degree of node i

$$\mathbf{T} = \begin{array}{c c} & \begin{array}{ccccccc} a & b & c & d & e & f & g \end{array} \\ \begin{array}{c} a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} & \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix} \end{array}$$

Notes:

Biased random walks in weighted graphs



random walker with initial state $X_0 = a$

we can use edge weights to **bias transition probabilities**, e.g.

$$T_{ij} := \frac{A_{ij}}{\sum_{k \in V} A_{ik}}$$

$$\mathbf{T} = \begin{array}{c c} & \begin{array}{ccccccc} a & b & c & d & e & f & g \end{array} \\ \begin{array}{c} a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} & \begin{bmatrix} 0 & \frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{8} & \frac{7}{8} & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{4}{7} & 0 & 0 & \frac{1}{7} & \frac{1}{7} & \frac{1}{7} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{bmatrix} \end{array}$$

Notes:

Stochastic matrices

- for any current state i, corresponding **row in transition matrix is probability mass function** capturing $P(X_{t+1} = j | X_t = i)$ for all states j
- rows in the transition matrix $\mathbf{T}$ of a Markov chain must sum to one, i.e.

$$\sum_{j \in V} T_{ij} = 1 (\forall i \in V)$$

- we call such a matrix **row-stochastic** or **right-stochastic**
- let $\pi = (\pi_i)_{i \in v}$ be **stochastic row vector**, i.e. $\sum_i \pi_i = 1$
- $\pi \cdot \mathbf{T}$ exists and is again a stochastic row vector

$$\mathbf{T} = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} \begin{array}{c} \begin{array}{ccccccc} a & b & c & d & e & f & g \end{array} \\ \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix} \end{array}$$

example

$$\pi = (1, 0, 0, 0, 0, 0, 0)$$

$$\pi \cdot \mathbf{T} = \left(0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0\right)$$

Notes:

Visitation probabilities

- let $\pi^{(t)} = \left(\pi_1^{(t)}, \pi_2^{(t)}, \ldots\right)$ be **probabilities of states** (visitation probabilities) at time t
- we call $\pi^{(0)}$ **initial state** of the Markov chain

example

$$\pi^{(0)} = (1, 0, 0, 0, 0, 0, 0)$$

$$\pi^{(1)} = \left(0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0\right) = \pi^{(0)} \cdot \mathbf{T}$$

$$\pi^{(2)} = \left(\frac{5}{12}, \frac{1}{4}, \frac{1}{6}, \frac{1}{6}, 0, 0, 0\right)$$

$$= \pi^{(1)} \cdot \mathbf{T} = \pi^{(0)} \cdot \mathbf{T}^2$$

$$\mathbf{T} = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} \begin{array}{c} \begin{array}{ccccccc} a & b & c & d & e & f & g \end{array} \\ \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix} \end{array}$$

using initial state $\pi^{(0)}$ and transition matrix $\mathbf{T}$ we can calculate **state $\pi^{(t)}$ for any time t** as $\pi^{(t)} = \pi^{(0)} \cdot \mathbf{T}^t$

Notes:

Total variation distance

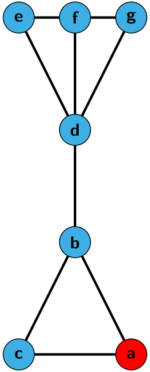
- evolution of state visitation probabilities of Markov chain resembles a **diffusion process**
- diffundere (Latin) = **to spread out**, to **distribute**
- how can we quantify the difference between two states π and π'?

total variation distance

For two probability mass functions π and π' the **total variation distance** between π and π' is defined as

$$\delta\left(\pi, \pi'\right) := \frac{1}{2} \sum_i \left|\pi_i - \pi'_i\right| \in [0, 1]$$

- we can use $\delta\left(\pi^{(t)}, \pi^{(t+1)}\right)$ to quantify the **evolution of states** from time t to $t+1$



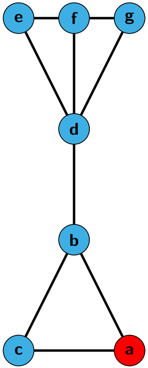
example

$$\pi^{(0)} = (1, 0, 0, 0, 0, 0, 0)$$
$$\pi^{(1)} = \left(0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0\right)$$
$$\pi^{(2)} = \left(\frac{5}{12}, \frac{1}{4}, \frac{1}{6}, \frac{1}{6}, 0, 0, 0\right)$$

Notes:

Group Exercise 14-01

- Consider a random walk in the graph on the right. For $\pi^{(0)} = (1, 0, \ldots)$ calculate the total variation distance between successive states $\pi^{(t)}$ and $\pi^{(t+1)}$ for different t.
- What do you observe in the evolution of $\delta(\pi^{(t)}, \pi^{(t+1)})$?



$$\mathbf{T} = \begin{array}{cc} & \begin{array}{ccccccc} a & b & c & d & e & f & g \end{array} \\ \begin{array}{c} a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} & \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix} \end{array}$$

Notes:

Stationary States of Markov chains

- in our example network, we observe

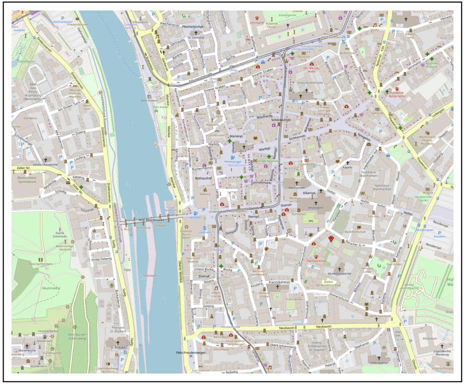
$$\delta\left(\pi^{(t)}, \pi^{(t+1)}\right) \to 0 \text{ for } t \to \infty$$

- in this example, Markov chain eventually reaches **stationary state**

$$\pi := \lim_{t \to} \pi^{(0)} \cdot \mathbf{T}^t$$

- after sufficiently long time, **it does not matter** at which time t we calculate visitation probabilities $\pi^{(t)}$

- allows us to make statements about **long-term behavior** of Markov chains (and random walks)



Map of Würzburg

image credit: Map created from openstreetmap data

Notes:

Stationary States and Eigenvalues

- Markov chain reaches **stationary state** $\pi := \lim_{t\to\infty} \pi^{(t)}$ such that

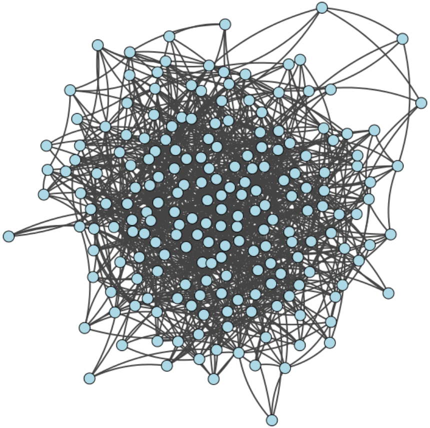
$$\pi = \pi \cdot \mathbf{T}$$

- this is an **eigenvalue problem** of the form

$$\pi \cdot \lambda = \pi \cdot \mathbf{T}$$

with eigenvalue $\lambda = 1$

- we can calculate stationary state as **left eigenvector of transition matrix corresponding to eigenvalue $\lambda = 1$**



convergence to stationary state for random walk in example graph

Notes:

Practice Session

- we show how we can simulate random walks in graphs using `python`
- we calculate the total variation distance between stochastic vectors

13-01 - Random Walks in Graphs

June 12 2023
Ingo Scholtes

In the first notebook we explore random walks in graphs. Random walk models have important applications in machine learning and recommender systems. In the first notebook, we show how we can calculate transition matrices and visitation probabilities of random walks. We use the python package `pathpy` to simulate and visualize random walks in graphs.

```python
import pathpy as pp
import numpy as np

import seaborn as sns
import matplotlib.pyplot as plt

from numpy import linalg as npl
import scipy as sp

plt.style.use('default')
sns.set_style("whitegrid")
```

Python

Transition matrices

We first write a function that computes a (row- or right-stochastic) transition matrix of a random walk process for a given network. The method should work for weighted, unweighted, directed, and undirected networks represented by a `pathpy.Network` object. For every `pathpy` network we can calculate the adjacency matrix, which is the basis for the definition of the transition matrix.

```python
n = pp.Network(directed=True)
n.add_edge('a', 'b', weight=2)
n.add_edge('a', 'c', weight=1)
n.add_edge('b', 'c', weight=1)
n.add_edge('b', 'a', weight=2)
n.add_edge('c', 'a', weight=5)
n.plot(edge_color='grey')

n.adjacency_matrix(n).todense()
```

Python

practice session

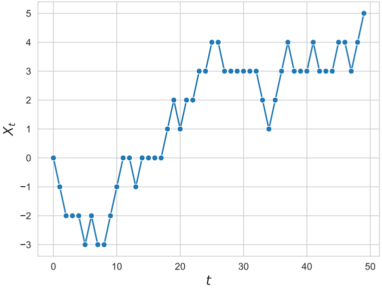
see notebooks 14-01 and 14-02 in `gitlab` repository at

→ `https://gitlab.informatik.uni-wuerzburg.de/ml4nets_notebooks/2025_sose_akids2`

Notes:

In summary

- we introduced **Markov chain models** for discrete-time categorical sequences
- key property of Markov processes is that they are **memoryless**, i.e. next state only depends on current state (and not on earlier steps in the trajectory)
- we can use Markov chains with **state space expansion** to model processes with any **finite memory** → self-study questions
- Markov chains are important class of probabilistic graphical models in **data science, machine learning, and theoretical physics**
- given transition matrix, how can we calculate **stationary state** of a Markov chain?



discrete-time stochastic process with **discrete state space**

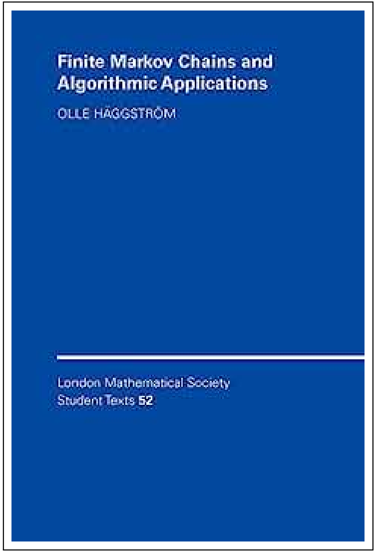
Notes:

Self-study questions

1. What is a stochastic process?
2. What is the Markov property and what is a Markov chain?
3. Consider a stochastic process, where we have a dependency of the next state to the current and the previous state, i.e. $P(X_{t+1} = s_{t+1} | X_t = s_t, X_{t-1} = s_{t-1})$. Can you change the state space such that you can use a Markov chain to model this process?
4. How can we define the transition matrix of a Markov chain capturing a random walk in an undirected graph?
5. Consider a random walk model, in which the random walker can ‘restart‘ in an arbitrary node in each step with a fixed probability. For the example graph shown on slide 8, compute the transition matrix of such a random walk with restart probability $\alpha = 0.5$.
6. Does a random walk in a graph with symmetric adjacency matrix have a symmetric transition matrix?
7. What do you get if you multiply a row-stochastic matrix from the right to a stochastic row vector π, i.e. $\pi \cdot \mathbf{T}$?
8. How is total variation distance defined?
9. Give an example for two probability vectors where the total variation distance is maximal.
10. For a right-stochastic transition matrix $\mathbf{T}$ and an initial distribution $\pi^{(0)}$, how can we compute visitation probabilities of a random walk at time t?

Notes:

References



reading list

- AA Markov: **Extension of the limit theorems of probability theory to a sum of variables connected in a chain**, reprinted in Appendix B of *R. Howard: Dynamic Probabilistic Systems, volume 1: Markov Chains*, 1971
- O Häggström: **Finite Markov Chains and Algorithmic Applications**, 2002

Notes: